

Bayesian Fine-Mapping with BayesC

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GWAS Regions Rarely Identify a Unique Causal Variant

- A GWAS association usually identifies an **associated region**, not a unique causal variant.
- Because nearby SNPs are correlated through **linkage disequilibrium (LD)**, several SNPs can show strong association even when only one is causal.

Fine-mapping moves from

associated region $\longrightarrow$ small set of plausible causal variants.

The goal is to quantify which variants remain plausible causal explanations after accounting for LD.

Fine-Mapping Connects Association Signals to Biology

- **Biological interpretation:** connect associated variants to genes, regulatory elements, and molecular mechanisms.
- **Functional follow-up:** prioritize variants, genes, and pathways for experimental studies.
- **Quantifying uncertainty:** distinguish one strongly supported variant from several LD-linked alternatives.
- **Functional information:** combine genetic evidence with eQTL, chromatin, conservation, and coding annotations.

Standard GWAS Tests One SNP at a Time

In a standard **genome-wide association study (GWAS)**, each SNP is tested separately for association with the phenotype.

For SNP j , fit a separate regression:

$$y_i = \mu + x_{ij}\beta_j + e_i, \quad e_i \sim \mathcal{N}(0, \sigma^2).$$

Test whether the SNP is associated with the phenotype:

$$H_0 : \beta_j = 0 \quad \text{versus} \quad H_1 : \beta_j \neq 0.$$

Each SNP is tested **one at a time**.

Standard GWAS Reports Marginal Evidence

For every SNP, GWAS typically reports:

- estimated effect $\hat{\beta}_j$
- standard error $\text{SE}(\hat{\beta}_j)$
- z - or t -statistic
- p -value

The result is a marker-by-marker summary of **marginal association evidence**.

Important: GWAS identifies **associated SNPs**, not necessarily the causal SNP.

LD Makes Multiple SNPs Look Associated

Nearby SNPs are often correlated because of **linkage disequilibrium (LD)**. If SNP j^* is causal,

$$y = x_{j^*}\beta_{j^*} + e.$$

Another SNP j can still show strong marginal association when

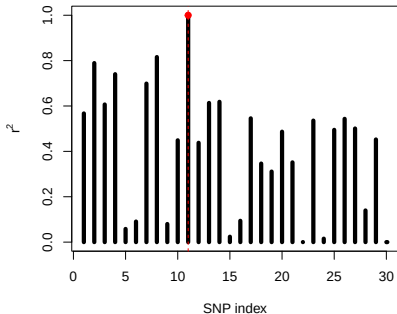
$$\text{Cor}(x_j, x_{j^*}) \neq 0 \quad \text{but} \quad \beta_j = 0.$$

A GWAS peak can therefore represent an **associated region**, not a unique causal variant.

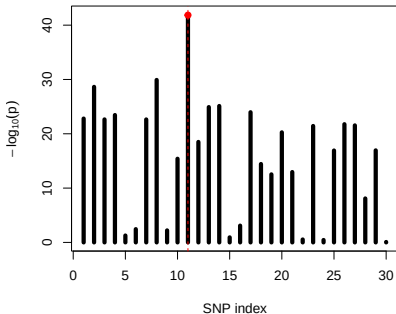
LD Explains the Cluster of Marginal Associations

The same simulated 30-SNP region is shown in both panels. The causal SNP is marked in red.

LD with the causal SNP



Marginal GWAS evidence



Fine-Mapping: Key Inputs

Fine-mapping aims to identify which SNPs within a GWAS-associated region remain plausible causal variants after accounting for LD.

- **Marker-level association evidence:** e.g. marginal effect estimates $\hat{\beta}_j$ and their standard errors $\text{SE}(\hat{\beta}_j)$.
- **LD between candidate SNPs:** represented by the LD correlation matrix, e.g. $R = \text{cor}(X)$.
- **A sparse joint model for genetic effects:** e.g. BayesC / spike-and-slab.

Here we use individual-level genotypes and phenotypes. Summary-statistic methods instead combine compatible GWAS evidence with external LD.

Fine-Mapping: Key Outputs

- **Posterior inclusion probabilities (PIPs):** probability that each SNP has a non-zero effect.
- **Posterior effect estimates:** model-averaged SNP effects.
- **Prioritized variant sets:** small sets of plausible causal variants.
- **Fine-mapping uncertainty:** one leading SNP or several LD-linked alternatives.

Fine-Mapping Fits SNP Effects Jointly

All SNPs are fitted **jointly**, allowing correlated SNPs to compete to explain the same association signal:

$$y_i = \mu + \sum_{j=1}^m x_{ij} b_j + e_i, \quad e_i \sim \mathcal{N}(0, \sigma_e^2).$$

- y — phenotype
- μ — overall mean
- X — genotype matrix
- b — SNP effects
- e — residual effects
- σ_e^2 — residual variance

BayesC Uses Inclusion Indicators

BayesC assumes that each SNP is either **inactive** or **active**. The point mass at zero creates a **sparse model** in which many SNP effects are exactly zero.

Introduce an inclusion indicator:

$$d_j \in \{0, 1\}, \quad d_j \sim \text{Bernoulli}(\pi).$$

Conditional on the indicator:

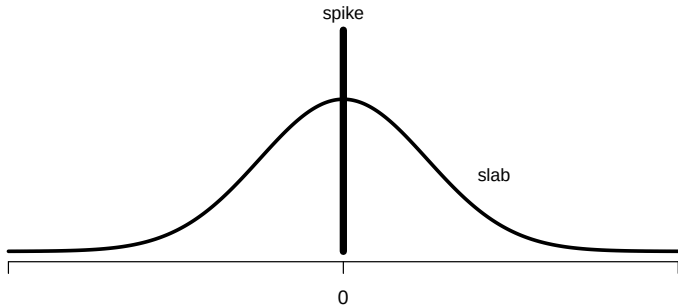
$$b_j \mid (d_j = 0) = 0,$$

$$b_j \mid (d_j = 1) \sim \mathcal{N}(0, \sigma_b^2).$$

BayesC Is a Spike-and-Slab Prior

Equivalently,

$$b_j \sim (1 - \pi)\delta_0 + \pi\mathcal{N}(0, \sigma_b^2).$$



π controls how often SNPs are active; σ_b^2 controls the variance of non-zero effects.

Correlated SNPs Compete Across Posterior Models

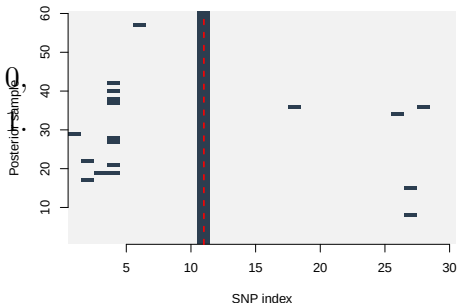
The sampler repeatedly explores possible explanations of the association signal.

Marker state at iteration t

$$b_j^{(t)} = \begin{cases} 0, \\ \text{non-zero draw,} \end{cases} \quad d_j^{(t)} = \begin{cases} 0, \\ \text{non-zero draw,} \end{cases}$$

The sampler updates d_j and b_j conditional on the current states of the other SNPs.

Sampled inclusion states



Posterior Samples Become Marker-Level Evidence

Posterior inclusion probability

$$\text{PIP}_j = \Pr(d_j = 1 \mid \mathcal{D}),$$

$$\widehat{\text{PIP}}_j = \frac{1}{T} \sum_{t=1}^T d_j^{(t)}.$$

PIP: how often SNP j is needed in posterior models.

Posterior mean effect

$$\hat{b}_j = E(b_j \mid \mathcal{D}),$$

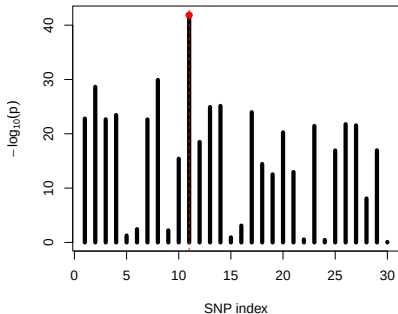
$$\hat{b}_j = \frac{1}{T} \sum_{t=1}^T b_j^{(t)}.$$

Posterior mean: the effect after averaging across posterior models.

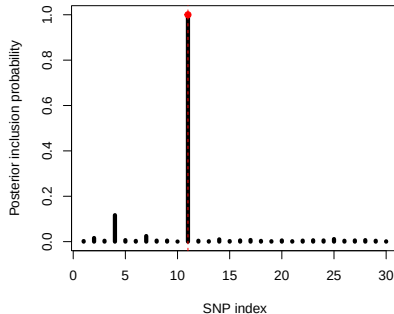
Joint Modelling Redistributes the GWAS Signal

The marginal GWAS signal is compared with support from the joint BayesC model.

Marginal GWAS



BayesC PIPs



PIP and Effect Size Answer Different Questions

Posterior inclusion probability

$$\text{PIP}_j = \Pr(b_j \neq 0 \mid \mathcal{D}).$$

Measures how often SNP j is included in posterior models.

A SNP can have a **high PIP but a moderate effect** if it is frequently active with a relatively small effect.

Posterior mean effect

$$E(b_j \mid \mathcal{D}).$$

Measures the model-averaged effect of SNP j , including iterations in which its effect is zero.

A SNP can have a **lower PIP but large non-zero effects** when it is selected.

Constructing a Local 0.9 Credible Set

A practical **local credible set (LCS)** summarizes the leading signal as a small set of plausible causal variants.

1. Identify the SNP with the highest PIP.
2. Select its LD neighborhood, for example $r^2 \geq 0.5$.
3. Rank these SNPs by decreasing PIP.
4. Add their PIPs until

$$\sum_{j \in \mathcal{C}} \text{PIP}_j \geq 0.90.$$

Interpreting the Local Credible Set

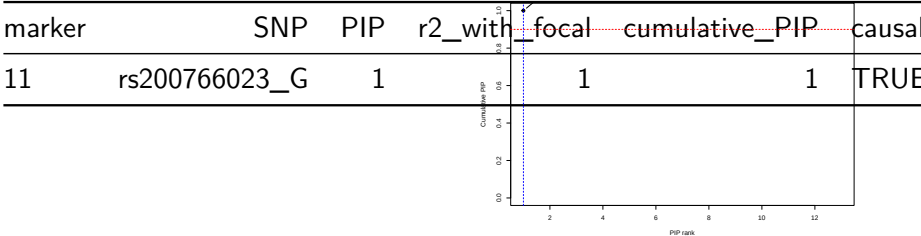
- PIPs come from the **joint model** and account for LD.
- A small LCS indicates a well-localized signal.
- A larger LCS indicates uncertainty among correlated SNPs.
- Raw PIPs are used without normalization.

Because BayesC allows multiple active SNPs, this is a **practical local summary**, not a formal 90% posterior coverage set.

Fine-Mapping Result: Local 0.9 Credible Set

Variants in the LCS

Cumulative PIP



The LCS contains 1 SNPs and includes the simulated causal SNP:
yes.

Functional Information Can Refine Fine-Mapping

When LD leaves several plausible SNPs, functional information can refine their prior support.

Functional information

Examples include:

- coding consequences;
- regulatory and eQTL evidence;
- chromatin accessibility;
- conservation and tissue- or cell-specific annotations.

Bayesian interpretation

Bayesian updating combines:

- genetic evidence from the locus;
- prior information for each SNP.

It changes a SNP's **prior plausibility**. It does not declare that SNP to be causal.

Annotation-Informed Inclusion Probabilities

Standard BayesC gives every SNP the same prior probability of being active:

$$d_j \sim \text{Bernoulli}(\pi).$$

Annotations can make that probability **SNP-specific**:

$$P(d_j = 1) = \pi_j.$$

A relevant annotation may therefore raise a SNP's prior probability of having a non-zero effect.

Annotation-Informed Effect Variances

Standard BayesC uses a common variance for active effects:

$$b_j \mid d_j = 1 \sim \mathcal{N}(0, \sigma_b^2).$$

Annotations can make the active-effect variance **SNP-specific**:

$$b_j \mid d_j = 1 \sim \mathcal{N}(0, \sigma_{b,j}^2).$$

An annotated SNP may therefore be assigned a larger prior variance for its non-zero effect.

Annotations Reweight Competing Causal Explanations

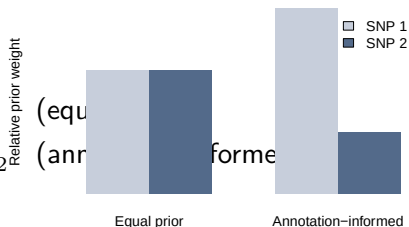
Suppose two SNPs have similar statistical evidence because they are in strong LD.

Equal versus
annotation-informed priors

$$P(d_1 = 1) = P(d_2 = 1) = \pi \quad (\text{equal prior})$$
$$P(d_1 = 1) = \pi_1, \quad P(d_2 = 1) = \pi_2 \quad (\text{annotation-informed prior})$$

where a relevant annotation can
imply $\pi_1 > \pi_2$.

Schematic prior reweighting



Annotation-Informed Fine-Mapping Workflow

Standard prior

- GWAS and LD evidence
- A common prior across SNPs
- Posterior PIPs

Annotation-informed prior

- GWAS and LD evidence
- SNP-specific prior information
- Annotation-informed PIPs

Both approaches report PIPs, ranked candidates, posterior effects, and local credible sets.

Only the prior assigned to each SNP changes—through its inclusion probability, non-zero effect variance, or both.

Summary

From association to joint evidence

- GWAS peaks identify **associated regions**, not unique causal variants.
- Joint fitting makes correlated variants compete to explain the signal.
- BayesC represents SNPs as inactive or active.

Interpreting the posterior

- PIPs and posterior means answer different questions.
- A BayesC LCS is practical, not automatically a formal 90% coverage set.
- Functional annotations modify priors, not the genetic evidence.

Further Reading

Companion tutorial: Bayesian Fine-Mapping with BayesC

- **Genome-wide fine-mapping — SBayesRC:** Wu Y, Zheng Z, Thibaut L, Lin T, Feng Q, Cheng H, Yengo L, Goddard ME, Wray NR, Visscher PM, Zeng J. Genome-wide fine-mapping improves identification of causal variants. *Nature Genetics* **58**, 940–951 (2026). DOI: 10.1038/s41588-026-02549-3
- **BayesC and BayesR for fine-mapping:** Shrestha M, Bai Z, Gholipourshahraki T, Hjelholt A, Rohde P, Fuglsang MK, Sørensen P. Enhanced genetic fine mapping accuracy with Bayesian Linear Regression models in diverse genetic architectures. *PLOS Genetics* **21**(7), e1011783 (2025). DOI: 10.1371/journal.pgen.1011783
- **Original SBayesRC method and prediction study:** Zheng Z et al. Leveraging functional genomic annotations and genome coverage to improve polygenic prediction of complex traits within and between ancestries. *Nature Genetics* **56**